Kai Barker

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ABOUT

Data Science senior at UC Santa Barbara with hands-on experience in statistical research, management consulting, and financial data analysis. Skilled in Python, R, SQL, and machine learning, with a track record of turning large complex datasets into clear decisions and measurable outcomes. Seeking a data analyst, data scientist, or analyst role where I can contribute immediately and grow.

EDUCATION

University of California, Santa Barbara — Santa Barbara, CA
Statistics and Data Science B.S. — Graduating June 2026
Major GPA: 3.86/4.0
Relevant Coursework: Machine Learning, Time Series, Stochastic Processes, Advanced Statistical Models, Computational Science, Differential Equations, Technology Management, Probability and Statistics, Problem Solving in Computer Science
Activities: Consult Your Community, Statistics & Data Science Club, Data Science Collaborative, Ski & Snow Club

EXPERIENCE

Engagement Manager — Santa Barbara, CA
Consult Your Community — February 2026 – Present

- Lead a team of 6 consultants on a 16-week EdTech engagement, defining 2 core project workstreams and serving as the primary liaison between client stakeholders and the analyst team.

Business Analyst — Santa Barbara, CA
Consult Your Community — October 2025 – Present

- Produced acquisition readiness assessment across operations, financials, and marketing for a small business client; identified 3+ areas to reduce owner dependency ahead of sale.
- Diagnosed over-reliance on owner relationships and gaps in client follow-up; recommended CRM and communication improvements projected to increase annual revenue by $40,000 and improve business attractiveness to buyers.
- Built 10+ Tableau visualizations to surface operational and financial gaps, using findings to drive CRM and SOP recommendations presented to client stakeholders.

Undergraduate Researcher — Santa Barbara, CA
Department of Statistics and Applied Probability — September 2025 – Present

- Analyze 70,000+ U.S. Senate stock trades in Python to test for statistically significant excess returns relative to market benchmarks.
- Calculate risk-adjusted performance using Sharpe ratios, finding members achieved 1.97 vs. market benchmark of 0.79
- Developed a data cleaning pipeline using FIFO matching and fuzzy matching to validate trade reporting accuracy, reconcile transaction records, and filter incomplete STOCK Act disclosures.

Research Assistant — Santa Barbara, CA
Foundations of Financial Technology Lab — September 2024 – Present

- Analyzed 20M+ Ethereum blocks and 100M+ wallets to quantify wealth distribution, calculated a Gini coefficient of 0.96.
- Produced visualizations tracking inequality evolution over 5 years, revealing the top 1% of wallets control 95%+ of Ether supply.
- Filtered institutional and Sybil addresses to isolate organic wallet behavior and improve the accuracy of distribution metrics.

PROJECTS

TrailForge AI Trail Recommendation Application — Santa Barbara, CA
SB Hacks XII Hackathon — January 2026 – January 2026

- Built a full-stack web app in 24 hours with OAuth 2.0, Python backend, and JavaScript/HTML frontend, integrating Strava, Google Search, and Gemini APIs for personalized trail recommendations.
- Engineered LLM prompts, designed a responsive UI with embedded trail links, deployed on Render within the hackathon window.

NeurIPS Ariel Data Challenge — Santa Barbara, CA
Competition - ESA Ariel Mission — July 2025 – September 2025

- Developed ensemble machine learning models (XGBoost + Gaussian Process Regression + Ridge) to predict exoplanet atmospheric compositions from 200+ GB of noisy spectral telescope data.
- Preprocessed 14,000 image files across 1,100+ exoplanets using signal processing and noise reduction techniques, achieving a GLL score of 0.311 across 283 wavelength channels.

MCMC & Simulated Annealing - Optimization Research — Santa Barbara, CA
Academic Research — May 2025 – June 2025

- Implemented and extended Metropolis-Hastings MCMC in Python to solve message decryption and the Traveling Salesman Problem, modeling both as permutation optimization over large discrete state spaces.
- Proposed a simulated annealing modification with an exponential cooling schedule, empirically validating a reduction in average decryption errors from 22.3 to 21.6 vs. baseline across 50,000 iterations.

SKILLS & INTERESTS

Languages: Python, R, SQL, JavaScript, TypeScript, Java, C++, HTML, CSS
ML & Statistics: Scikit-learn, XGBoost, Guassian Processes, SHAP, Feature Engineering, Statistical Modeling, Time Series
Data & Tools: Pandas, NumPy, Matplotlib, Tableau, Excel, Git, API Integration, LaTeX, Data Pipeline Development, PowerPoint
Interests: Basketball, Hiking, Backpacking, Rock Climbing, Cooking, Running, Pickleball